

RESEARCH ARTICLE

Blockchain-Enabled Zero Trust Architecture for Privacy-Preserving Cybersecurity in IoT Environments

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ABSTRACT This research introduced a new novel “Unified Quantum-Resilient Blockchain-Zero-Knowledge Proofs Privacy Authentication Framework (QBC-ZKPAF)” to upgrade the IoT environments with greater security. To enable privacy-preserving authentication, access control, and secure communication, the framework integrates blockchain technology with Zero Trust Architecture (ZTA) and post-quantum cryptography. A hybrid Reinforcement-Lattice Blockchain KeyGen for quantum-resilient key generation, Deep Q-Network Multi-Factor Secure Key (DQN-MFSK) for dynamic selection of keys, and Zero-Knowledge Proof for privacy-preserving signatures are employed to achieve secure IoT settings. This architecture entails data privacy and confidentiality, auditability and traceability, and withstanding evolving threats, including potential threats in terms of quantum attacks. It then uses blockchain technology for recording unalterable data of identity and access management while Zero-Knowledge Proofs (ZKP) ensures authentication and verification without revealing sensitive information. By decentralizing identity management and enabling multi-factor authentication, QBC-ZKPAF provides robust security and privacy solutions for IoT networks. The experimental results demonstrate the model's effectiveness with 98% privacy preservation, 700 TPS throughput, 0.7 J energy consumption, 0.98 quantum resilience, and 96% access control effectiveness, making it highly suitable for modern IoT and blockchain applications.

INDEX TERMS IoT security, privacy preservation, access control, blockchain technology, zero trust architecture, quantum resilience, data confidentiality, user authentication, decentralization, secure communication.

I. INTRODUCTION

Recently, IoT is one of the major technologies and bridging devices across healthcare, smart cities, and industrial automation. In IoT, cybersecurity is a significant challenge as the number of devices continues to grow, so do the inherent limitations in their computing and security capabilities. Generally, IoT environments are very dynamic and decentralized. In such an environment, the constant stream of sensitive data between connected devices makes it a prime target for cyberattacks like data breaches, malware intrusion, and denial of service (DoS) attacks [1]. Traditional security models, particularly those reliant on the notion of a secured

perimeter, are not applicable to IoT networks simply because most of the time, it is quite impossible to define any specific perimeter within which these devices will be working when they're in distributed, cloud-based, or edge environments [2]. This necessitates enhanced security architectures designed to address the specific vulnerabilities of IoT systems more effectively.

For this reason, ZTA has been found as one promising solution. Zero-Trust is a security model that assumes no entity, whether inside or outside the network, should be trusted by default. Every request for access or communication is treated as potentially harmful, requiring continuous authentication and authorization based on strict identity verification, even if it comes from within the trusted network perimeter. Moreover, ZTA is based on the simple principle of “never trust,

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always verify,” which means that no device, user, or application will be trusted by default regardless of whether they are inside or outside the network perimeter [3]. Every access request is viewed as potentially malicious and constantly verified by enforcing the least privilege principle [4]. It has been shown that ZTA was effective in thwarting the lateral movement of attackers and generally protecting the network [5]. Even though ZTA does improve security, it introduces new challenges, particularly regarding privacy preservation. The continuous authentication and access control processes in ZTA can inadvertently expose sensitive user data or device attributes, raising privacy concerns [6].

Currently, blockchain has received much attention with its decentralization and tamper-resistance properties that promise to help tackle a vast set of security and privacy challenges within IoT environments. Blockchain is a decentralized and distributed digital ledger technology that records transactions across many computers in a way that ensures security, transparency, and immutability. It uses consensus algorithms like Proof of Work (PoW) or Proof of Stake (PoS) to validate and add new blocks of data, making it resistant to tampering and fraud. Furthermore, blockchain works as a distributed ledger that stores data across many nodes. This ensures no single point of failure may develop and assures the integrity of the data [7]. Cryptographic techniques use blockchain in such a way that data is immutable and verifiable best suited to secure IoT networks [8]. However, blockchain’s transparency along with immutability poses a privacy risk since sensitive information may spill across every node [9]. Blockchain’s scalability along with resource requirements add up to be a concern to the IoT devices limited by computational power [10].

With these limitations, there is a need to have a hybrid approach that thrives on the advantages offered by ZTA along with leveraging the decentralized and tamper-resistant characteristics of blockchain. It suggests a blockchain-enabled ZTA to address both security and privacy challenges in IoT environments. To implement decentralized identity management with multi-factor authentication (MFA) within the ZTA framework to ensure robust authentication and access control without compromising privacy, an amalgamated approach is proposed. In this context, decentralized identity management provides secure storage of IoT device identities on a blockchain basis so that no central authority can control and compromise user data [11]. This erases all risks associated with central identity repositories, such as data breaches or insider threats.

In addition, it would utilize advanced cryptography techniques in addition to ring signatures and designated verifier signatures. These cryptographic techniques enhance the preservation of privacy during authentication and communication. Ring signature allows signing a message without revealing the identity of the signer and thus ensures the preservation of user anonymity while keeping the integrity of a message [12]. Designed for the verifier signatures, selective disclosure of the signer’s identity is possible; that is, only

the designated verifier can authenticate the signer’s authenticity [13]. Thus, these methods guarantee that while the system is able to trace and track harmful behaviour when it matters, user privacy prevails under normal conditions.

In addition, MFA is included in the ZTA framework, strengthening the access controls. MFA requires the authentication of users through at least one other form of independent authentication factor such as something they know, something they have, and something that they are. This limits even unauthorized access if one factor has been compromised [14]. The decentralized nature of blockchain means the MFA credentials are going to be stored and authenticated with tamper-proof authenticity that cannot be accessed by phishing attacks or credential stuffing attacks [15]. Thus, this paper aims to introduce a novel cybersecurity mechanism for IoT settings by integrating blockchain with ZTA and MFA through providing decentralized, and tamper-proof credential storage with enhanced quantum resilience. Quantum Resilience refers to the ability of a system or protocol to remain secure in the face of potential threats from quantum computers. Quantum computers have the potential to break current cryptographic algorithms, like RSA and ECC, which rely on the difficulty of factoring large numbers or solving discrete logarithms. Quantum resilience involves developing cryptographic techniques (quantum-safe or post-quantum cryptography) that can withstand the computational power of quantum machines. This combination minimizes the risk of unauthorized access even if one factor is compromised. Besides, it effectively guards against phishing and credential-stuffing attacks. It ensures both enhanced access control and privacy preservation for sensitive IoT environments. This blockchain-enabled zero-trust architecture can be applied to various IoT environments such as smart healthcare systems for secure patient data management [16], smart cities for enhanced traffic and energy management [17], and industrial IoT networks for robust device authentication and threat detection [18]. Blockchain is particularly suited to the QBC-ZKPAF architecture due to its inherent features of decentralization, immutability, and transparency. These features are essential for secure identity management and privacy preservation in IoT networks. Unlike other Distributed Ledger Technologies (DLTs), blockchain ensures that all recorded data such as access permissions and identity verification logs cannot be tampered with once added to the ledger and provides tamper-proof records. Additionally, its decentralized nature eliminates the need for a central authority. It reduces single points of failure and increases system robustness, which is critical in large-scale, distributed IoT environments. The main contribution of this content is as follows:

- To propose a unified framework that integrates blockchain, ZTA, and post-quantum cryptography to ensure robust, privacy-preserving authentication and access control in IoT environments while utilizing quantum-resistant zero-knowledge proof verification via QBC-ZKPAF framework.

- To ensure secure key generation and verification with post-quantum cryptographic methods by offering resilience against potential quantum computing threats, while preserving privacy and strengthening IoT security via quantum-resilient authentication.
- To introduce a hybrid approach combining Deep Q-Network (DQN) with multi-factor authentication by providing an adaptive, and robust method for selecting secure keys and enhancing the overall security of IoT systems via DQN-MFSK technique.

The upcoming section of the paper is arranged as follows; section II covers the literature survey, section III covers the scheme model, section IV covers proposed methodology, section V covers the results and discussion, and section 6 covers conclusion.

II. REVIEW OF SOTA ON ZTA

Liu et al. [19] proposed a novel blockchain-based information sharing in ZTA. This proposed approach was verified to be useful by performance evaluation on an Ethereum-based blockchain platform, and it was found to be secure in the context of universal composability. Nonetheless, this scheme suffers from communication latency as well as protocol overhead in calculation.

Thantharate and Thantharate [20] introduced ZeroTrust-Block a comprehensive blockchain-based platform that erases shortcomings in popular health IT systems for the secure and private management of health data. Hyperledger Fabric has clearly shown the efficacy of the design to enhance security, privacy, integrity, and interoperability for exchanging health data. However, it has scalability issues, the existing privacy policies, and practical hurdles regarding adoption.

Alevizos et al. [21] introduced an Intrusion Detection and Prevention System (BIDPS), enhancing ZTA on endpoints with Blockchain support. Two key objectives of the BIDPS are: (1) to identify and stop attacker strategies and tactics before they reach the lateral movement stage, according to the MITRE ATT&CK enterprise matrix; and (2) to remove trust from the endpoint itself and transfer it to the blockchain; an immutable system of explicit trust. One serious limitation that went with this research was the size of the BIDPS.

Han et al. [22] proposed building a ZTBDS on the 6G edge of IoT to collect and store data using the PoR technique based on updatable polynomial commitment instead of PoW. A better scheme to incorporate distributed storage was designed. In this design, the consensus algorithm was proposed to be the PoR. Subsequently, storing the data used the dynamic accumulator instead of the Merkle tree. However, there was a lack of privacy protection.

Li et al. [23] presented blockchain-based edge computing architecture for smart cities with a zero-trust basis. This method relies on the model of a digital identity and supports two-way authentication, fine-grained authorization, and trust management using distributed ledgers. However, it has scalability problems, and computational intensiveness increases,

especially when using this approach based on blockchain technology and consensus mechanisms.

Gai et al. [24] proposed a blockchain-based access scheme for the sharing of multi-organization data in a zero-trust setting. It established its trusted framework through a consortium blockchain and was supported by a Role-Based Access Control (RBAC) model implemented over a multi-signature protocol and smart contracts. However, it has disadvantages such as potential scalability as the number of organizations increases and technical management complexity, especially in a dynamic environment.

Chaudhry and Hydros [25] established a blockchain-based framework to ensure security in the banking sector through the integration of zero-trust security and consensus algorithms to secure transactions and thwart cyberattacks on bank resources. Its value-added propositions surround the ability to preserve the confidentiality and integrity of data through decentralization and immutability. Nevertheless, one limitation of this study was that the suggested model must be implemented in practice, and a trust algorithm must be designed. These ultimately become difficult issues within the practical application and utilization of this QBC-ZKPAF to effectively evaluate accesses in real-world scenarios.

Li et al. [26] proposed a security protection plan that relies on ShangMi cryptography (SM) techniques, zero trust, and blockchain technology. The plan offered the assurance of validity, non-repudiation, and tamper-proofness of keys and events since key management relies on Hyperledger Fabric while the storage of trust evaluation events. However cyber security issues exist in EV chargers and charging management cloud platforms due to a huge attack surface and lack of rigorous research.

Miraz and Ali [27] discussed the deployment of BC technology to increase the level of security and privacy for IoT. BC technology, with cryptography-based techniques and distributed consensus, provides an immutable ledger that eliminates the need for mediators, thus mitigating the issues of safety due to the growing number of IoT devices. However, no challenges specific to the implementation of blockchain technology in IoT environments were identified and solved in this study but instead need to be further researched and developed.

Edo et al. [28] designed a ZTA to improve security within patient healthcare data; this thereby answers vulnerabilities concerning insider threats, through data breaches, offering continuous authentication and context-based access policies, by way of preventing lateral movement in the systems. Although the ZTA has been successful in diverse arenas, complexity in implementation was the major limitation, especially when it was to be integrated with all healthcare systems and technologies such that effective data protection was ensured. In 2024, Padma and Ramaiah [29] proposed a unique blockchain-based system named SecPrivPreserve to guarantee the integrity and security of IoT data. Through a number of stages, including initialization, registration, data protection, authentication, data access control, validation,

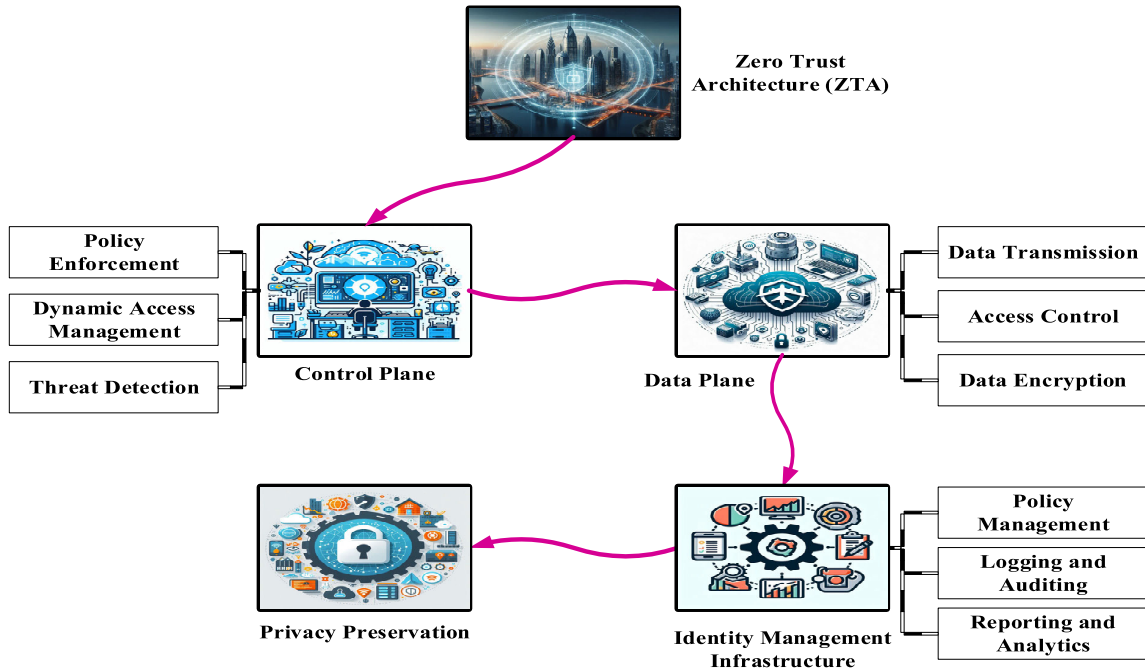


FIGURE 1. Architecture of the ZTA.

and data sharing and download, the suggested SecPrivP-reserve architecture guaranteed security. In 2023, Padma and Ramaiah [30] addressed an ensemble consensus based on trust and the Grey Wolf Optimization (GWO) method to raise network trust levels. For this process, Proof of Work (PoW) and Proof of Stake (PoS) consensus processes were used by the trust-based ensemble consensus. In 2023, Chinnasamy et al. [31] suggested a reliable access control system that made use of smart contracts to increase security while exchanging electronic health information between different patients and medical professionals. Moreover, this idea provided a proactive solution for safe data exchange in mobile computing while shielding private health data from any threats. In 2021, Gangwani et al. [32] provided a directed acyclic graph, a novel distributed ledger solution for the IoT. IOTA provides a data transmission protocol called masked authenticated messaging for safe data exchange between IoT devices, overcoming the blockchain's present drawbacks. In 2023, Zhao et al [33] offered a Merkel-tree-based technique to connect the condensed data on the public blockchain with the raw sensing data held off-chain. Through this technique, the raw sensing data kept off-chain was granted the data immutability attribute of a public blockchain. Besides, a self-protection approach that grouped the raw sensing data into chained blocks with a reasonable quantity of PoW was used to safeguard the raw sensing data kept off-chain.

1) PROBLEM STATEMENT

From the above study it is clear that in [19] communication latency and protocol overhead in calculations hinder efficiency, in [20] scalability issues and practical hurdles limit widespread adoption, in [21] the size of the BIDPS poses

significant operational challenges, in [22] lack of privacy protection undermines overall security in the proposed system, in [23] scalability problems and increased computational intensity with blockchain usage, in [24] potential scalability issues and technical complexity in dynamic environments, in [25] practical implementation and trust algorithm design pose significant challenges, in [26] cybersecurity issues in EV chargers create vulnerabilities in the proposed plan, in [27] no specific challenges of blockchain in IoT were identified or addressed, in [28] implementation complexity in healthcare systems complicates effective data protection. In [29], the SecPrivPreserve framework failed to address scalability and computational overhead challenges. In [30], the complexity of integrating sidechains and optimizing PoW/PoS increased system overhead and implementation difficulty. The complexity of implementing and maintaining secure smart contracts exposed challenges in scalability and flexibility in real-world healthcare environments [31]. A drawback of the IOTA-based approach is its potential scalability issues and lack of widespread adoption compared to traditional blockchain systems, which may hinder its integration into existing IoT infrastructures [32]. Furthermore, a drawback in [33] is the reliance on public blockchain networks, which can introduce latency and scalability challenges when handling large volumes of real-time sensing data.

III. ZERO-TRUST ARCHITECTURE (ZTA)

Normally, ZTA [34] is a security framework that requires all users, whether they are in or outside the organizational network, to be continuously authenticated, authorized, and validated before they gain access to network applications and data. Figure 1 shows the architecture of the ZTA. The control

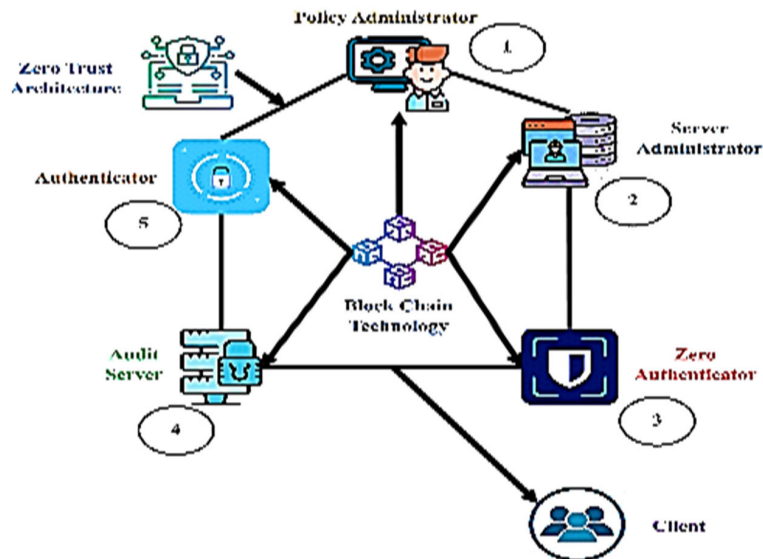


FIGURE 2. Proposed scheme model.

plane, data plane, and identity management infrastructure comprise the three layers of the ZT network.

2) CONTROL PLANE

The control plane includes the policy administrator and policy engine, which check and validate the requests that a user makes to obtain information. Users get authentication tickets from the ticket authorisation server. Logs get uploaded through the log server.

3) DATA PLANE

This is determined by the data plane executor when it receives the result of the authentication of the policy administrator; in other words, when a client requests access to the service resources of the system.

4) IDENTITY MANAGEMENT INFRASTRUCTURE

It is a necessary part of the identity management infrastructure of ZTA, which includes identity management services, device management services, registration authority, certification authority, key generation centre, and audit services [35]. This research work is focused on Identity management infrastructure in the case of privacy-preservation on the authentication.

A. SYSTEM MODEL

The proposed privacy-preserving authentication scheme includes four main stakeholders: Client, Policy Administrator, Server Administrator, and Audit Server. This proposed privacy-preserving authentication scheme combines Blockchain Technology with ZTA and guarantees safe, traceable, and effective authentication while still maintaining user privacy. The architecture of the scheme model is represented in Figure 2.

1) CLIENT

The client is the data requestor, starting with a request to access. The client holds an authentication ticket that is issued from a trusted signer; for example, a certificate authority, and converts that original ticket into a designated ticket by relying on some privacy-preserving protocol, such as ZKP. The designated ticket is then forwarded to the Server Administrator but sent under privacy conditions so the ticket information is not exposed inappropriately.

2) POLICY ADMINISTRATOR

This user is a decision-maker with control over the ZTA's policy decisions. ZTA ensures that access to resources is granted based on explicit authentication and authorization, rather than network location or trust assumptions. Every access request, whether it originates from an internal or external entity, must be verified before being granted. The authentication of the client's access request by the Policy Administrator is then cross-checked against a defined set of access policies stored on the blockchain. Assuming the access request has passed through the authentication gateway, the Policy Administrator issues an authentication ticket that contains cryptographic signatures to ensure integrity and authenticity of access.

3) SERVER ADMINISTRATOR

It manages the data resources in the server. After having the designated ticket from a client, it will verify the validity of the ticket using the signatures that were provided by the Policy Administrator. Once verified properly with a valid designated ticket, then the Server Administrator will allow the client to access the requested service resources. Similarly, all such transactions along with access grants are maintained over blockchain so as to have auditability and traceability.

4) AUDIT SERVER

The Audit Server is a tracing authority that conducts any type of inquiry into possible troubles related to designated tickets. Once the Server Administrator realizes that something is amiss with a designated ticket, they seek arbitration from the Audit Server. Now, tracing keys are used by the Audit Server to identify and trace who issued such problematic designated tickets.

B. SECURITY REQUIREMENTS

These are the main security requirements for this work that ensure robustness, confidentiality, and integrity for privacy-preserving authentication and access control in IoT environments through the integration of Blockchain Technology with ZTA. Figure 3 provides the structure of security requirements.

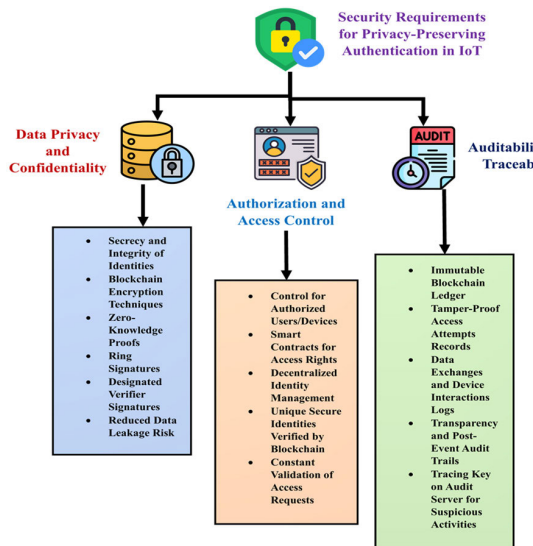


FIGURE 3. Security requirements.

1) DATA PRIVACY AND CONFIDENTIALITY

This framework makes use of a combination of blockchain encryption techniques combined with ZKP to keep data private. It uses ring signatures and designated verifier signatures to provide anonymity while permitting to verification of access rights claims.

2) AUTHORIZATION AND ACCESS CONTROL

They have to access the IoT environment with the control for authorized users or devices. Smart contracts on a blockchain enforce pre-defined access rights and policies. Decentralized identity management is used so that unique, secure identities verified through blockchain are maintained by the users. Blockchain-based access tokens ensure authorized user access to the resources, and this is evident from Figure 4. Here, decentralized systems exposed increased security in terms of robustness to various attacks, privacy with respect to data confidentiality, and performance in terms of throughput.

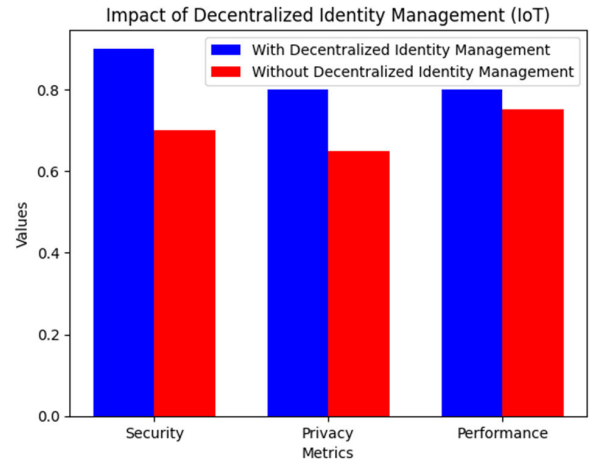


FIGURE 4. Impact of decentralized identity management on QBC-ZKPAF for security, privacy, and performance metrics.

Implementing ZTA, all access requests have to be validated constantly. Identity management solutions that are decentralized have a positive impact on privacy (increasing values by at least 0.25) and security (adding 0.15) respectively. Thus, users are empowered to take ownership of their identifiers while minimizing the risks existing in the centralized binding of users to their identifiers where risks of a single point of failure are present.

3) AUDITABILITY AND TRACEABILITY

Since the blockchain ledger is immutable, all access attempts, data exchanges, and device interactions are maintained in a tamper-proof way that supports transparency and post-event audit trails. In cases of suspicious activities or breaches, tracing down to the source is supported through the tracing key on the audit server in ZTA.

C. PROPOSED QBC-ZKPAF MODEL

This research proposed a QBC-ZKPAF that incorporates blockchain technology and ZTA for privacy-preserving authentication and access control in IoT environments. ZKP [36] is used to protect user data during authentication without revealing sensitive information. In the context of proving identity P to verifier V without disclosing the actual information, the ZKP proof follows the properties in Eq. (1), in which y denotes the outcome of a cryptographic function f that P knows, but does not disclose directly. Only the proof of knowing x without disclosing x itself is shared, preserving privacy.

$$P : \text{Prove } x \text{ such that } f(x) = y \quad (1)$$

Ring signatures are applied to ensure user anonymity in transactions. A ring signature [37] allows any member of a defined group to sign a message on behalf of the group without revealing which specific member signed it as shown in Eq. (2), in which S indicates the ring signature, m signifies

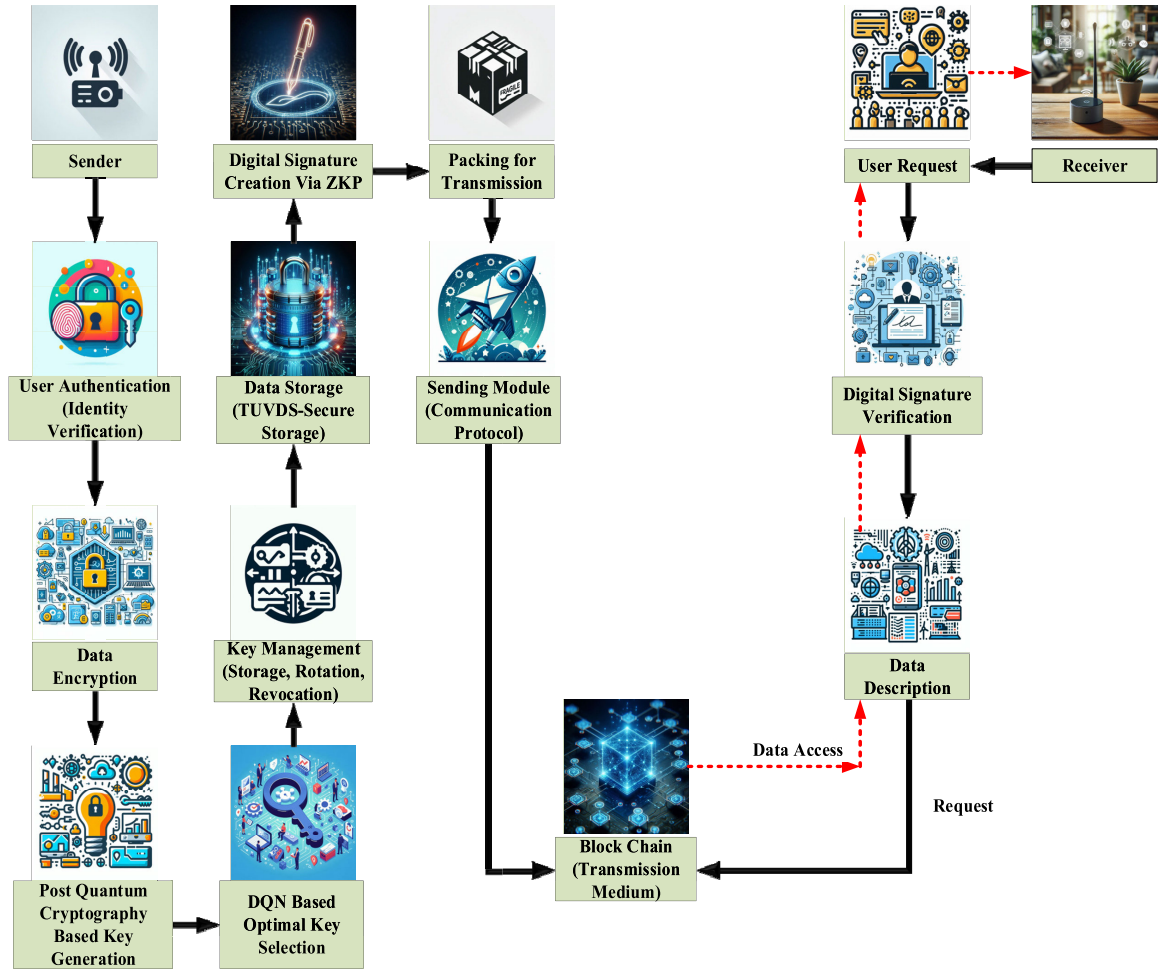


FIGURE 5. Architecture of proposed ZTA model based on Unified Quantum-Resilient Blockchain-ZKP Privacy Authentication Framework (QBC-ZKPAF).

the message, and $\{PK_1, PK_2, \dots, PK_n\}$ refers to the public keys of group members. The verifier confirms that someone in the group signed m , but not which member, preserving individual privacy.

$$S = \sigma(m \{PK_1, PK_2, \dots, PK_n\}) \quad (2)$$

To achieve scalable consensus in a decentralized manner, the framework employs Delegated Proof of Stake (DPoS) [38], where stakeholders vote for delegates who then validate transactions. Let v_i represent the votes received by a delegate i . The probability p_i of i being selected as a block producer is proportional to Eq. (3).

$$p_i = \frac{v_i}{\sum_j v_j} \quad (3)$$

This mechanism ensures efficient consensus without compromising scalability, as only elected delegates participate directly in block production.

MFA [39] key generation and authentication process relies on combining multiple factors such as password-based key $K_{password}$, biometric-based key $K_{biometric}$, and token-based

key K_{token} . The final authentication key K_{auth} is generated as stated in Eq. (4), in which H points to a secure hashing function, combining multiple factors to enhance security.

$$K_{auth} = H(K_{password} || K_{biometric} || K_{token}) \quad (4)$$

The architecture of the proposed scheme is described in Figure 5. It shows the requesting data from the receiver side and the accessed data from the sender side.

D. SETUP

System parameters are predefined in the setup phase. The cryptographic protocols to be used for the system are also chosen. These include post-quantum cryptographic algorithms, optimizing the size of the key, and blockchain infrastructure. Combine post-quantum cryptography and blockchain to set up a quantum-resilient identity infrastructure. Use lattice-based signatures such as CRYSTALS-Dilithium to sign public keys when registering them for identity management against quantum attacks. Blockchain ensures, beyond that, immutable records for

public keys, so any adversary could never tamper with any form of identity registration.

E. KEY GENERATION

A new hybrid technique, Reinforcement-Lattice Blockchain KeyGen (RLBK) introduced here using Reinforcement Learning (RL) [40] with postquantum lattice-based cryptography [41] toward secure purposes for key generation. The RL algorithm optimizes the generation of keys by the dynamical response to attack vectors for improvement of resilience. In lattice-based cryptography, security is derived from the hardness of lattice problems, which are believed to be resistant to attacks by quantum computers. The keys are generated based on a lattice structure, ensuring post-quantum security. Let $L \subset R^n$ represent a lattice generated by a basis $B = \{b_1, b_2, \dots, b_n\}$ where each $b_i \in R^n$. Using the Short Integer Solution (SIS) problem, lattice-based cryptographic schemes generate a public-private key pair (pk, sk) . Given a matrix A and an error vector e , the public key pk is computed as given in Eq. (5), where sk states the private key, and the error vector e ensures security by making it difficult to recover sk from pk .

$$pk = A \cdot sk + e \quad (5)$$

Here, RL is used to optimize the key generation process by adjusting security parameters based on changing attack vectors. The RL agent observes the environment's state, s_t , which includes data on potential attack vectors and security requirements. The RL agent selects actions A_t from a set of possible adjustments to key generation parameters such as the lattice dimension n , and error distribution. The reward r_t is designed to improve security and efficiency, rewarding the agent for selecting parameters that maximize resilience to attacks as defined in Eq. (6), where $f(\text{security}, \text{efficiency})$ addresses a function balancing security and computational efficiency, and α states weighting factor penalizing configurations vulnerable to certain attacks.

$$r_t = f(\text{security}, \text{efficiency}) - \alpha \cdot \text{attack vulnerability} \quad (6)$$

Using a policy $\pi(s_t, A_t)$, the RL agent updates its action selection strategy over time, optimizing key generation for resilience against evolving threats. Once the RL-optimized lattice-based keys are generated, they are securely stored on a blockchain to ensure integrity, traceability, and immutability. The generated key pk (public key) is stored in a blockchain transaction, providing a permanent, tamper-proof record. Each key pk is hashed using a cryptographic hash function H , and the hash is stored as a blockchain transaction T as illustrated in Eq. (7).

$$T = \text{blockchain} \cdot \text{Append}(H(pk)) \quad (7)$$

The blockchain structure ensures that any modification attempt on the stored key would invalidate the hash, guaranteeing tamper resistance. Since blockchain transactions are immutable, the stored keys cannot be altered or deleted,

ensuring that each key is traceable and verifiable over time. Combining these components, the RLBK framework is represented via the generated key pk stored on the blockchain as portrayed in Eq. (8), in which pk points to lattice-based public key optimized through RL, and T addresses blockchain transaction holding the hash of pk , ensuring integrity and immutability.

$$T = \text{blockchain} \cdot \text{Append}(H(pk)), \text{ where } pk = A \cdot sk + e \quad (8)$$

F. KEY SELECTION

A new hybrid technique, DQN-MFSK, merges DQN [42] with multi-factor authentication to robustly select and generate a key. Here, using DQN, keys are selectively accessed through various factors such as a User Password Key obtained from the password K_{password} , Biometric Key $K_{\text{biometric}}$ derived from personal biometric data like fingerprints, and Token Key K_{token} , which is issued from a physical security token. Real-time adaptation in DQN algorithms exploits this in combination with the environmental security factors, thereby optimizing these keys and improving their security features. The environment state includes factors related to the security level, threat level, and the quality of each authentication factor. For example, if a high-security environment is detected, the DQN will prioritize stronger authentication factors. Besides, the DQN selects actions A_t from a set of choices, which correspond to different combinations of the multi-factor keys $K_{\text{password}}, K_{\text{biometric}}, K_{\text{token}}$. The DQN optimizes the choice of these keys based on the security requirements of the current environment. The reward function incentivizes the selection of combinations that maximize security and efficiency. It assigns higher rewards to actions that produce stronger keys and penalizes those that produce weaker keys. The selected factors are combined based on the DQN policy to generate the overall key K . The combined key K is a function of the selected factors as shown in Eq. (9), in which g points to encryption or hashing function that merges the individual keys into a single secure key.

$$K = g(K_{\text{password}}, K_{\text{biometric}}, K_{\text{token}}) \quad (9)$$

The DQN algorithm dynamically adapts the selection of $K_{\text{password}}, K_{\text{biometric}}, K_{\text{token}}$ based on real-time feedback from the environment, optimizing k for maximum security. The DQN continually learns and updates its policy $\pi(s_t, A_t)$ to select the best combination of multi-factor keys, improving security and privacy. The Q-value for each state-action pair is updated using the Bellman equation as given in Eq. (10), in which α states learning rate, γ signifies discount factor, r_t means reward at time t , s_{t+1} denotes next state, and A' addresses best action for that state.

$$Q(s_t, A_t) = Q(s_t, A_t) + \alpha(r_t + \gamma Q(s_{t+1}, A') - Q(s_t, A_t)) \quad (10)$$

The analysis of the contributions of each of the techniques employed in the QBC-ZKPAF is based on a “with

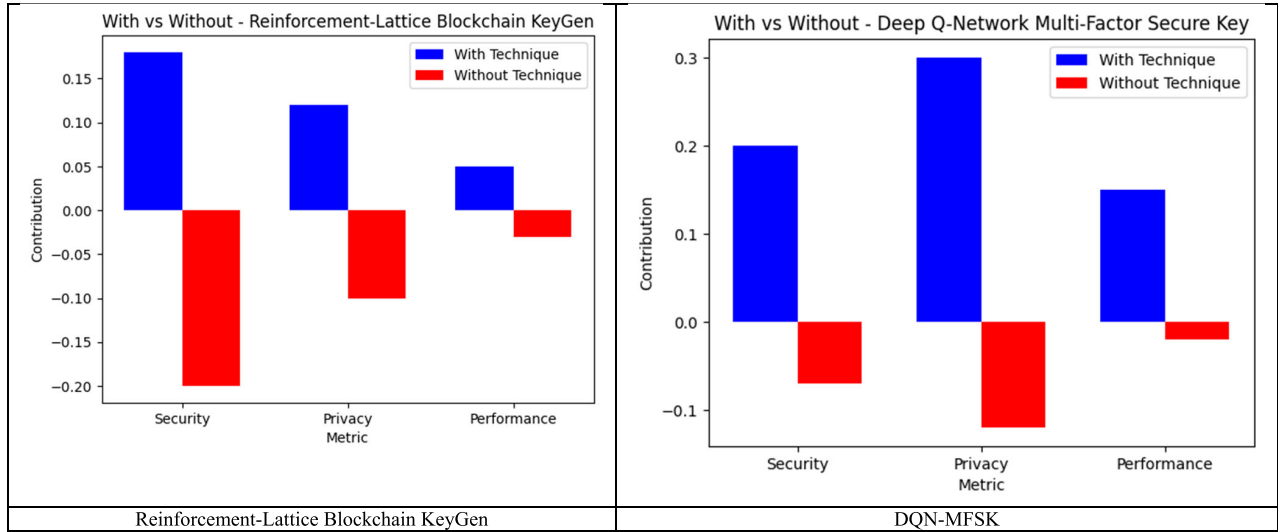


FIGURE 6. Impact of RL on QBC-ZKPAF for security, privacy, and performance.

and without” assessment is shown in Figure 6. The Reinforcement-Lattice Blockchain KeyGen Adds primarily to security (+0.25) and privacy (+0.15). This technique is of utmost importance in quantum-secure key generation and shields the system from any quantum-based crypto attacks. Furthermore, DQN-MFSK improves privacy (+0.30) and security (+0.20). By utilizing deep RL for dynamic key selection, the system can respond efficiently and choose the most secure key appropriate for the situation which enhances access control and privacy protection in changing IoT ecosystems. The final secure key K generated by DQN-MFSK is represented as shown in Eq. (11), where each component $K_{password}^{DQN}$, $K_{biometric}^{DQN}$, K_{token}^{DQN} is selected dynamically by the DQN based on environmental factors, optimizing both privacy and security

$$K = g(K_{password}^{DQN}, K_{biometric}^{DQN}, K_{token}^{DQN}) \quad (11)$$

G. SIGNING

In the signing phase, a signature is produced from the private key of the user in order to authenticate or confirm the message. The process includes the property of ZKP to achieve privacy along with its assurance of transaction authenticity. Introduced UDVS by adopting ZKP with post-quantum cryptography. In this way, users can generate signatures verifiable without access to sensitive information about the transaction, hence enhancing the user’s privacy. Then the user or an IoT device signs a data message such as a transaction or a request using their private key (sk) for Data (D) to sign the document and the signature is calculated as defined in Eq. (12).

$$Sig = Sign(sk, D) \quad (12)$$

H. PUBLIC VERIFICATION

Public verification allows any party with public access to the key to verify validity of the signature. In the case of the

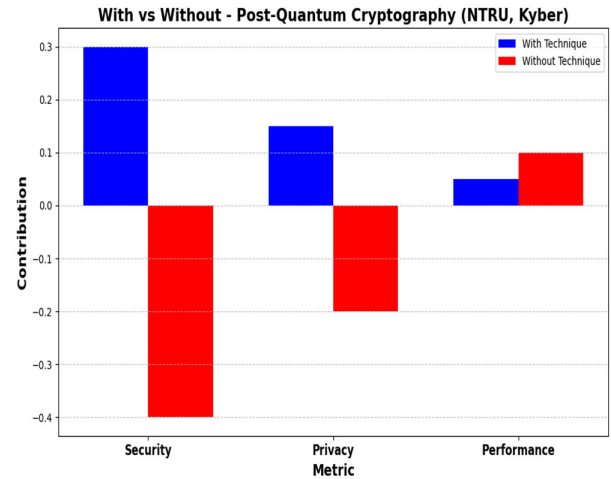


FIGURE 7. Impact of post-quantum cryptography on QBC-ZKPAF for security, privacy, and performance.

ZT network, that’s a necessity of course, as the identities are authenticated securely without revealing the information unnecessarily. Combine post-quantum signature verification with a blockchain-backed public key lookup for integrity. Adding a ZKP privacy layer ensures the signing information is private while still being on-chain verified. It is a signature verification that the other nodes verify the signed data using the corresponding public key as shown in Eq. (13).

$$Verify(pk, Sig, D) \Rightarrow True/False \quad (13)$$

Figure 7 ensures the highest security boost (+0.30) to withstand quantum threats. NTRU KIS AKJHA and Kyber KIS AKJHA guarantee, apart from other measures, that the engravings at the foundation of the system’s architecture will not be shaken even by the inventions of quantum computing.

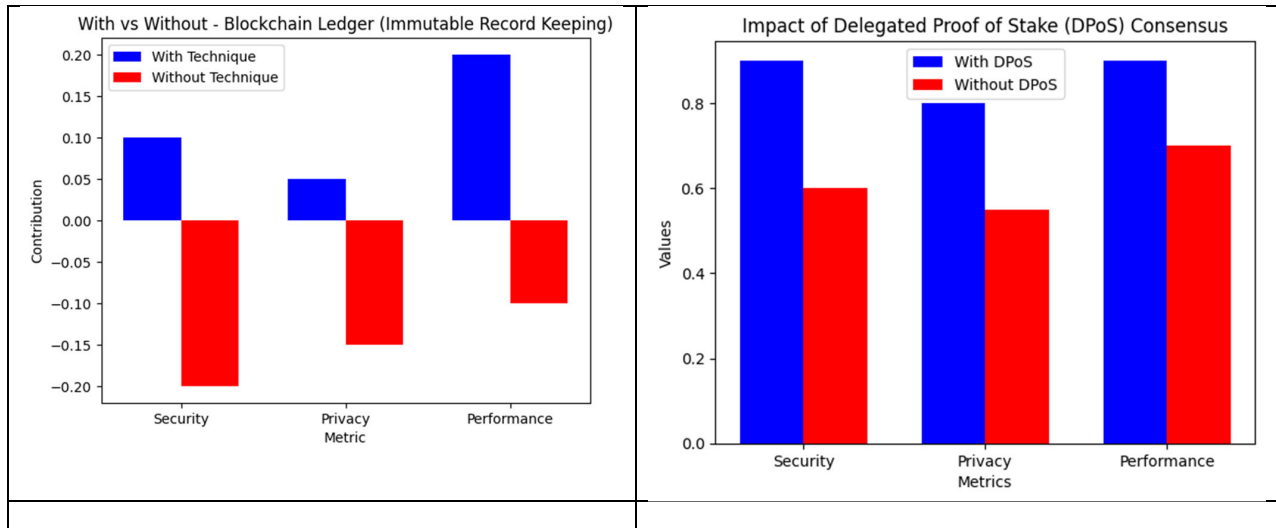


FIGURE 8. Impact of blockchain on QBC-ZKPAF for security, privacy, and performance metrics.

I. DESIGNATION

In this phase, the signature is assigned to a specific verifier to whom only the designated verifier is allowed to authenticate the signature. It incorporates a dynamic key agreement protocol, such as post-quantum New Hope. Therefore, a secure key exchange is ensured between the user and the designated verifier. The public key of the designated verifier is also stored in the blockchain for privacy and verifiability without third-party exposure.

1) DESIGNATION VERIFICATION VIA ZERO-KNOWLEDGE PROOF

At this step, the verifier is able to validate the signature. In the privacy-preserving scheme, the verification needs to ensure that the signature is authentic and, at no point, it has to leak any sensitive information about who signed or what transaction was done. Use ZKP during the verification of designated verification to show that the verifier indeed verified the signature correctly without revealing the contents of the signature or the identity of the signer. Such a verification ensures post-quantum security while it preserves privacy. In this process, the IoT device or service verifies the proof π to ensure the user possesses the necessary rights without knowing the user's identity as stated in Eq. (14).

$$\text{VerifyZKP}(pk, \pi) \Rightarrow \text{True/False} \quad (14)$$

J. BLOCKCHAIN-BASED TRANSMISSION

The signed data is passed on to the blockchain, meaning that its record will be irreversible:

- Store data and signatures in a transaction.
- Put it in a block and add it to the blockchain. There is an infinite amount of these blocks.

The impact of blockchain on QBC-ZKPAF in terms of privacy, security and performance is shown in Figure 8. The aspect of blockchain ledger (records that cannot be altered in

any manner) enhances security (+0.10) and privacy (+0.05) to a moderate extent. Since the access logs, transactions, and identity verification processes within the blockchain are all within the scope of the technology, such activities are tamper-proof and no information is ever lost.

The consensus mechanism, DPoS is used in the proposed approach, as it directly impacts transaction speed, scalability, and energy efficiency in decentralized blockchain systems. While DPoS is efficient, its trade-off is the potential concentration of power among the selected delegates. The parameter values of blockchain are shown in Table 1.

K. TRACING

Tracing is an important function in privacy-preserving systems so that the real signer is identified under certain conditions, such as lawfully required for an investigation or fraud detection. Combine blockchain with ZKP to implement traceability. With a legal requirement, then the blockchain provides an audit trail. Then, with a reveal based on ZKP, the tracing authority demonstrated that tracing was authorized while keeping unnecessary information private. It does thus balance both privacy and accountability.

As per Figure 9, the QBC-ZKPAF designed with blockchain integration and zero trust principles, promises secure communication and user privacy in IoT environments. The control scheme employs dynamic key generation, zero-knowledge proof, and blockchain technology for data transmission.

Algorithm 1 defines the pseudocode of proposed QBC-ZKPAF.

IV. RESULTS AND DISCUSSION

The proposed Unified Quantum-Resilient Blockchain-ZKP Privacy Authentication Framework, QBC-ZKPAF, will thus integrate blockchain and principles of zero-trust architecture. The QBC-ZKPAF has been implemented in the

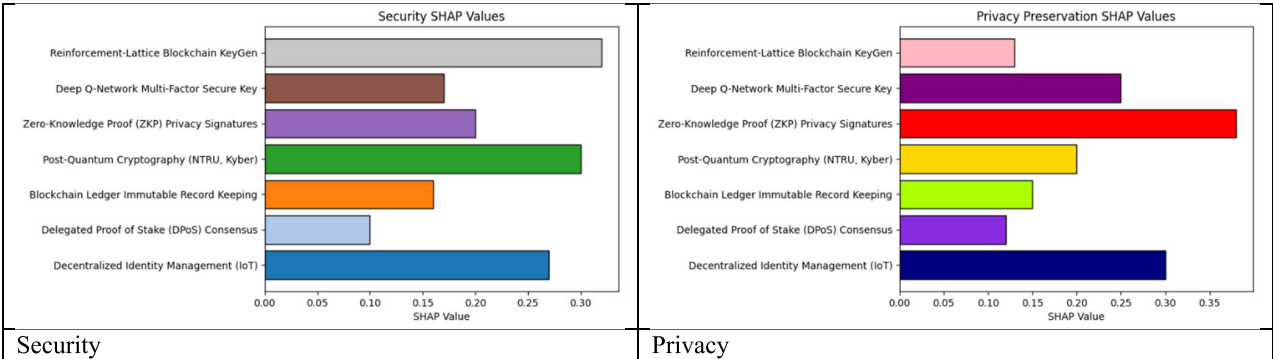


FIGURE 9. SHAP analysis on QBC-ZKPAF in terms of security, privacy.

TABLE 1. Parameter values of blockchain.

Category	Component	Details/Values
Hardware	Storage	1 TB SSD for high-speed transaction recording
	RAM	32 GB DDR4
	Power Consumption	0.7 J per transaction (energy efficiency)
	Network Bandwidth	10 Gbps Ethernet for handling high throughput (700 TPS)
	GPU	Nvidia Tesla V100 (for accelerating ZKP and post-quantum cryptography computations)
Software	CPU	8-core Intel Xeon or AMD EPYC (2.5 GHz or higher)
	ZKP Protocol	zk-SNARKs (Zero-Knowledge Succinct Non-Interactive Argument of Knowledge)
	OS	Linux
	Multi-Factor Authentication	DQN for dynamic key selection
	Key Generation Algorithm	Reinforcement-Lattice Blockchain KeyGen
Blockchain Metrics	Database	LevelDB for storing blockchain states and transactions
	Cryptographic Algorithms	Post-Quantum Lattice-Based Encryption (NTRU, Kyber)
	Consensus Mechanism	DPoS
	Blockchain Platform	Hyperledger Fabric (customized for post-quantum resilience)
	Transaction Throughput	700 TPS
Other Parameters	Quantum Resilience	0.98
	Privacy Preservation	98%
	Auditability & Traceability	Full traceability and immutable record-keeping
	Access Control Effectiveness	96%
	Network Latency	< 50 ms
	Block Size	1 MB (adjustable based on IoT data requirements)

Python platform, and it has been validated using the Kaggle database: Edge-IIoTset Cyber Security Dataset of IoT & IIoT in <https://www.kaggle.com/datasets/mohamedamineferrag/edgeiiotset-cyber-security-dataset-of-iot-iiot?select=Edge-IIoTset+dataset>. This evaluation enhances security in IoT to a considerable extent with strong authentication, access control, and communication schemes while preserving user privacy-a couple of deal-breakers for managing data and devices in IoT networks. This dataset helps analyze net-

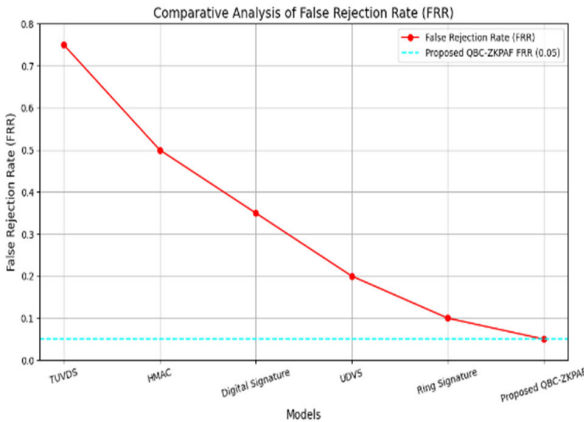


FIGURE 10. Analysis of authentication time over SOTA.

work traffic patterns, detect anomalies, and identify access behaviours that inform policy enforcement and security decisions. Besides, suspicious access patterns, facilitate improved decision-making by the Policy and Server Administrators and support the auditing and traceability functions of the Audit Server. This data is given as input to RL, based on this traffic data, the RL model transmits data and generates secured keys. The QBC-ZKPAF is compared with the other existing techniques like TUVDS [43], HMAC [44], Digital signature [45], UDVS [46] and Ring Signature [47].

Figure 10 shows the Authentication Time of the QBC-ZKPAF compared to other existing techniques. The Authentication Time of the QBC-ZKPAF is 100 ms, whereas the Authentication Time of the TUVDS, HMAC, Digital signature, UDVS and Ring Signature is 800 ms, 500 ms, 600 ms, 700 ms and 400 ms, respectively. There is a significant improvement in the QBC-ZKPAF. Especially in applications where speed is critical, such as in high-frequency trading or time-sensitive security operations, this improvement could offer a substantial competitive advantage.

The throughput of the proposed model is measured based on Transactions Per Second (TPS). A transaction refers to an action such as data submission, authentication request, or message exchange within the system. It includes:

Algorithm 1 Pseudocode of Proposed QBC-ZKPAF

Step 1	Authentication Algorithm with Privacy Preservation Apply ZKP - Client generates proof using ZKP that they know the required identity credentials without revealing them. - The verifier checks the proof and grants access based on the policy validation via Eq. (1)
Step 2	Ring Signature for Anonymity - The client signs the authentication request with a ring signature, which includes multiple public keys via Eq. (2). - The policy administrator checks the validity of the ring signature to authenticate without knowing the exact identity.
Step 3	Blockchain Integration for Auditability - Each time a user requests access, a transaction is created and recorded in the blockchain. - The blockchain stores the hash of the authentication ticket, ensuring tamper-proof records of access attempts. - In case of suspicious activities, the audit server traces the origin of the request using the immutable blockchain ledger.
Step 4	DPoS for Consensus - Stakeholders vote for delegates who are responsible for validating transactions using Eq. (3). - Delegates produce blocks based on the votes they receive, optimizing blockchain scalability and performance.
Step 5	MFA and Key Generation - Combine multiple authentication factors: $K_{password}$, $K_{biometric}$, K_{token} - Use a cryptographic function to merge them into a final authentication key as per Eq. (4). - RL optimizes this process based on the attack vectors detected, ensuring robustness against evolving threats.
Step 6	RLBK for Key Generation - Use lattice-based cryptography for key generation. The key pair (pk, sk) is derived based on the lattice dimension and security parameters via Eq. (5). - RLBK observes the security environment and optimizes the key generation parameters based on rewards for resilience against attacks via Eq. (6).
Step 7	Blockchain Transaction for Key Integrity - The generated key pk is stored on the blockchain after being hashed. - The blockchain transaction provides tamper-proof storage for keys, ensuring they cannot be altered or deleted without detection using Eq. (8).
Step 8	Dynamic Key Selection via DQN - Observe the security environment and evaluate key factors: $K_{password}$, $K_{biometric}$, K_{token}. - Use DQN to select the most appropriate combination of keys based on the current environment, optimizing for security via Eq. (9). - The DQN continually learns from feedback to adapt the selection process over time using Eq. (11).

Step 9	Apply signing to produce signature as per Eq. (12)
Step 10	Public Verification Add a ZKP privacy layer to ensure signing data
Step 11	Blockchain-Based Transmission - Store data and signatures in a transaction. - Put it in a block and add it to the blockchain. There is an infinite amount of these blocks.

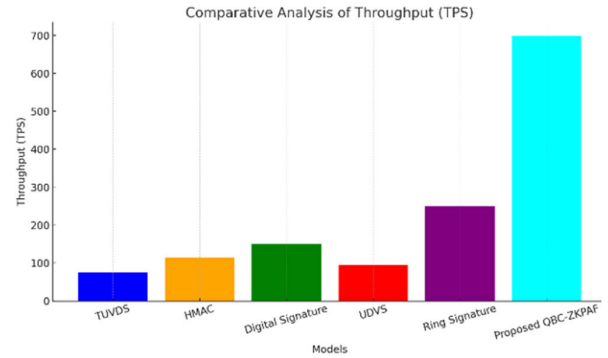


FIGURE 11. Throughput (TPS) of the QBC-ZKPAF over SOTA.

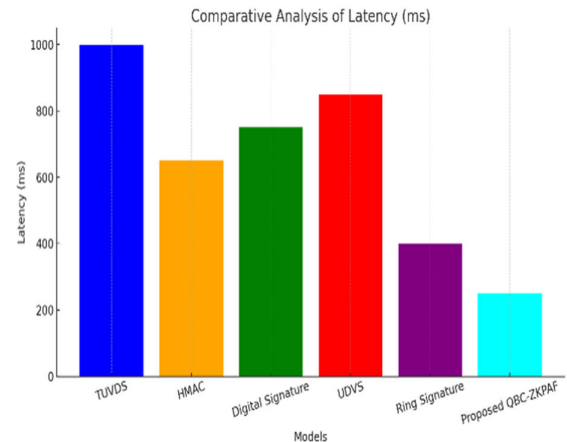


FIGURE 12. Latency of the QBC-ZKPAF over SOTA.

- Data exchanges: Secure transmission of data between devices.
- Blockchain operations: Adding, verifying, or querying data in the blockchain ledger.
- ZKP processes: The verification of Zero-Knowledge Proofs to validate identities without revealing sensitive information.

Figure 11 shows the Throughput comparison of the proposed methodology with others currently in practice in the presented model, 700 TPS is reached where 50 TPS is achieved by TUVDS, HMAC reaches 80 TPS, 100 TPS by a digital signature, 70 TPS by UDVS and lastly 200 TPS by ring signature.

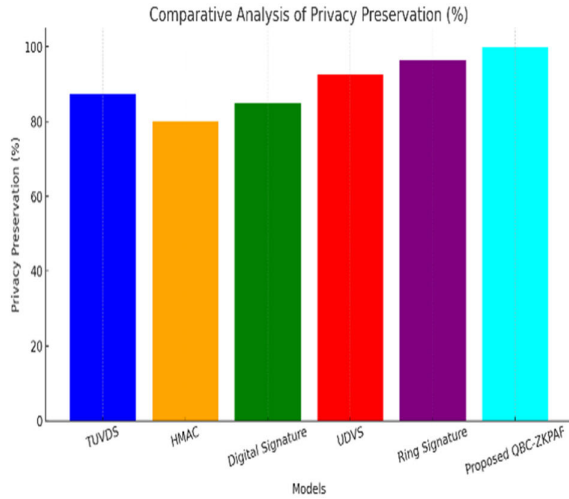


FIGURE 13. Privacy preservation of the QBC-ZKPAF over SOTA.

Figure 12 shows the comparison of Latency Achieved by the Proposed Methodology against Other Approaches Currently in Use in the Model. The QBC-ZKPAF achieves 200 ms of latency whereas the TUVDS is 1000 ms; HMAC is 600 ms, 700 ms by using digital signature, 800 ms by using UDVS, and 400 ms by using ring signature. It is a much better value.

Privacy preservation is evaluated via a measure ZKP efficiency. In a ZKP system, the prover can convince the verifier that they know a piece of information without revealing the information itself. In order to evaluate how effectively ZKP is used in the system, metrics such as the time taken to generate and verify proofs as well as the security level (resistance to attacks) are used to assess its privacy-preserving PP capabilities. Let $P_{attack\ success}$ denote the adversary's success probability in breaching the system. Including ZKP, which reduces the adversary's information advantage without revealing underlying secrets, this framework adjusts the adversary's success probability to include a ZKP-based reduction factor Z . In systems resistant to attacks, privacy preservation is often measured in terms of the adversary's success probability as indicated in Eq. (15), where $P_{attack\ success}$ specifies probability that an adversary successfully breaches the system.

$$PP = 1 - P_{attack\ success} \quad (15)$$

With ZKP, the adversary's success probability is reduced as in Eq. (16).

$$P_{attack\ success} = P_{baseline-attack-success} \cdot (1 - Z) \quad (16)$$

Here, $P_{baseline-attack-success}$ is the initial probability of attack success without ZKP measures and Z is a factor between 0 and 1 representing the effectiveness of ZKP in mitigating the adversary's advantage. A higher Z value corresponds to greater reduction in attack success probability.

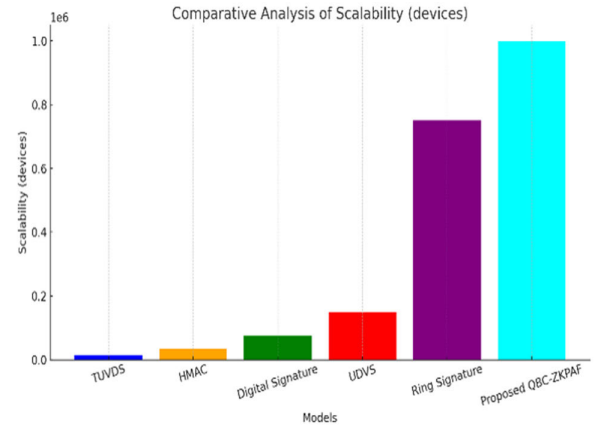


FIGURE 14. Scalability of the QBC-ZKPAF over SOTA.

Substituting Equation (16) into Equation (15), the privacy preservation (PP) by incorporating the impact of ZKP is determined in Eq. (17).

$$PP = 1 - (P_{baseline-attack-success} \cdot (1 - Z)) \quad (17)$$

Figure 13 shows the privacy preservation of the proposed methodology compared to other approaches currently in use in the model. The proposed approach achieves 98% Privacy Preservation as compared with 85% for TUVDS, 78% for HMAC, 82% for digital signatures, 90% for UDVS, and 94% for ring signatures.

In the proposed framework, system's scalability is achieved by applying efficient data handling, customized consensus mechanism, distributed ledger management, and support for large-scale identity and authentication. With decentralized identity management and MFA, the system securely authenticates and manages a growing number of users or devices without overloading any single component. If N represents the number of users or devices and T denotes throughput, the scalability of a system incorporating Zero-Knowledge Proofs (ZKP) for privacy-preserving authentication and operation validation is expressed using Eq. (18)

$$S = \frac{T_{new}}{T_{initial}} \times \frac{N_{initial}}{N_{new}} \times Z \quad (18)$$

Here, $T_{initial}$ denotes the throughput at the initial load, T_{new} is the throughput at the increased load, $N_{initial}$ represents the initial number of users or operations, N_{new} is the increased number of users or operations, and Z denotes the Zero-Knowledge Proof efficiency factor, which accounts for the reduced computational overhead and enhanced privacy. This factor improves the scalability when ZKP reduces verification complexity and latency.

Figure 14 shows the scalability of the proposed methodology compared with other methods that are presently in use within the model. As seen in Figure 9, the proposed methodology obtained 0.98×10^6 Scalability compared to 0.02×10^6 for TUVDS, 0.03×10^6 for HMAC, 0.08×10^6

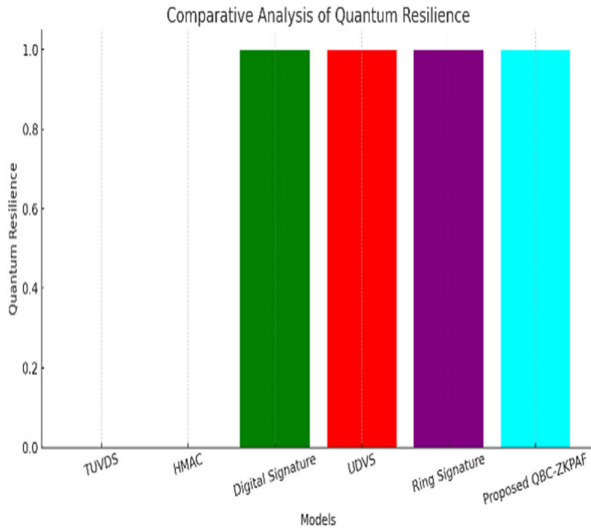


FIGURE 15. Quantum resilience of the QBC-ZKPAF over SOTA.

for digital signatures, 0.12×10^6 for UDVS, and 0.8×10^6 for ring signatures.

Quantum resilience measures a system's ability to resist quantum computing attacks, particularly on cryptographic protocols. It is defined as the ability to maintain security strength against quantum adversaries. It is mathematically defined in Eq. (17), where classical vs. Quantum attack success analyzes how quickly a quantum algorithm reduces the cryptographic key's strength compared to classical algorithms, and security level in bits measures the effective security bits the system retains when facing quantum attacks. Quantum resilience (QR) measures a system's ability to maintain security in the presence of quantum attacks, relative to classical security strength. The enhanced security level, incorporating ZKP, is represented as follows in Eq. (19).

$$QR_{ZKP} = \frac{\text{Quantum Security Level with ZKP}}{\text{Classical Security Level}} \quad (19)$$

Here, Quantum Security Level with ZKP (QSL_{ZKP}) is the effective security level when employing cryptographic protocols resistant to quantum attacks, augmented by ZKP-based mechanisms. Classical Security Level (CSL) denotes the security strength in bits under classical cryptographic analysis. Finally, ZKP Contribution represents the added robustness from zero-knowledge proof techniques, defined by a security improvement factor Δ_{ZKP} , which enhances privacy and integrity by enabling verification without revealing sensitive data.

Initially, QSL_{ZKP} to explicitly include ZKP's contribution in Eq. (20).

$$QSL_{ZKP} = QSL_{base} + \Delta_{ZKP} \quad (20)$$

Here, QSL_{base} is the baseline quantum security level without ZKP. Thus, the final Eq. (21) is determined for the enhanced

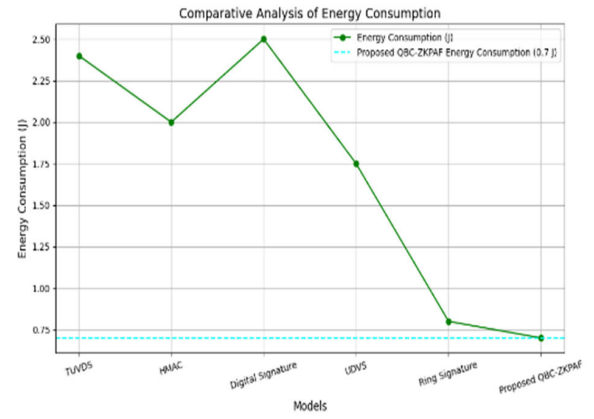


FIGURE 16. Energy consumption of the QBC-ZKPAF over SOTA.

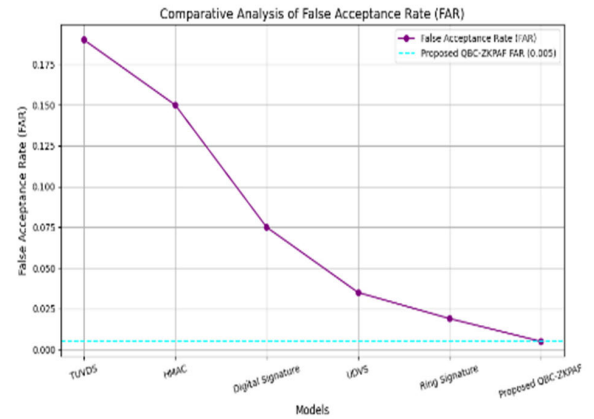


FIGURE 17. False Acceptance Rate (FAR) of the QBC-ZKPAF over SOTA.

security level, incorporating ZKP

$$QR_{ZKP} = \frac{QSL_{base} + \Delta_{ZKP}}{CSL} \quad (21)$$

Figure 15 displays the comparison of the quantum resilience that the proposed methodology could provide with other approaches currently used inside the model. The presented method provides 0.98 of quantum resilience compared to TUVDS at 0.01, HMAC at 0.03, digital signatures at 0.97, UDVS at 0.97, and ring signatures at 0.97.

In Figure 16, the energy consumption that the proposed approach provides is compared with other methods currently deployed inside the model. Comparing it with TUVDS at 2.4 J, HMAC at 2 J, digital signatures at 2.5 J, UDVS at 1.75 J, and ring signatures at 0.8 J, the proposed approach that is being given uses only 0.7 J of energy. That is very much more valuable.

Figure 17 shows the comparison of the FAR that has been achieved by the proposed approach with other methods that are currently implemented within the model. The proposed method requires only 0.005 energy compared to TUVDS at 0.19, HMAC at 0.15, digital signatures at 0.075, UDVS at 0.035 and ring signatures at 0.019.

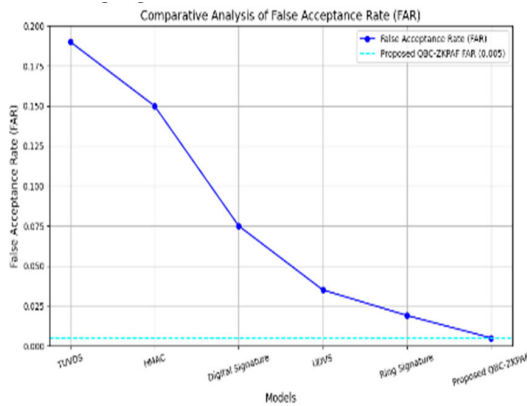


FIGURE 18. False Rejection Rate (FRR) of the QBC-ZKPAF over SOTA.

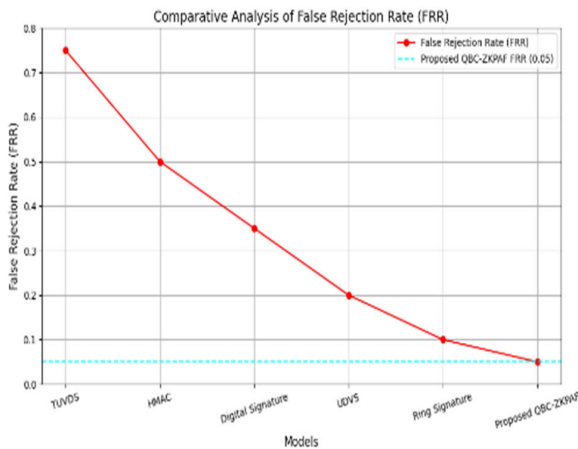


FIGURE 19. Access control effectiveness of the QBC-ZKPAF over SOTA.

Figure 18 shows a comparison of the achieved FRR of the proposed approach with several techniques that have now been adopted within the model. As compared to TUVDS at 0.75, HMAC at 0.5, digital signatures at 0.35, UDVS at 0.2, and ring signatures at 0.1, it is clear that this proposed approach above uses only 0.05 energy.

The achieved Access Control Effectiveness of the proposed approach is compared with a number of approaches already incorporated into the model shown in Figure 19. It is deduced that the above-proposed approach employs 96% of Access Control Effectiveness compared to TUVDS at 90%, HMAC at 90%, digital signatures at 92%, UDVS at 93%, and ring signatures at 94%.

The evaluation of potential quantum attack risk outcomes shown in Fig. 20 on the QBC-ZKPAF system discloses important weaknesses that affect the security, privacy, and efficiency of IoT networks. On the one hand, conventional public key infrastructures (PKI), which are susceptible to quantum computer algorithms, indicate a high level of risk. However, while this means longer keys are needed for safety reasons, symmetric key cryptography also has its drawbacks. As a result, developing post-quantum cryptography alongside ZKP provides a solution to overcoming those hurdles, however, the cost of remediation is rather high.

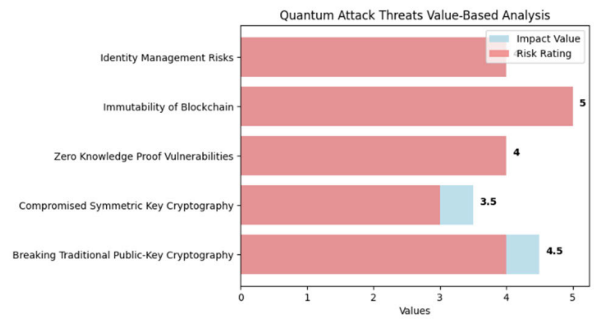


FIGURE 20. Evaluation of potential quantum attack risks on the QBC-ZKPAF system.

In order to measure the overhead for overall transactions in blockchain-enabled ZKP networks, the overall throughput overhead is measured by measuring the number of transactions the system can process per second (TPS) with and without blockchain/ZKP in Eq. (22).

$$\text{Throughput Overhead} = \frac{\left(\frac{\text{TPS without blockchain}}{\text{ZKP}} - \frac{\text{TPS with blockchain}}{\text{ZKP}} \right)}{\frac{\text{TPS without blockchain}}{\text{ZKP}}} \quad (22)$$

The computational overhead proposed model is evaluated by measuring the time taken for tasks like ZKP generation and verification or for adding a transaction to the blockchain as given in Eq. (23).

$$\text{Processing Time Overhead} = \frac{\left(\frac{\text{Time taken with blockchain}}{\text{ZKP}} - \text{Base time} \left(\frac{\text{without blockchain}}{\text{ZKP}} \right) \right)}{\text{Base time}} \quad (23)$$

Table 2 compares the robustness of the proposed Quantum Blockchain-based Zero-Knowledge Proof Authentication Framework (QBC-ZKPAF) against various potential attacks in comparison to other models like TUVDS, HMAC, Digital Signature, UDVS, and Ring Signature. The proposed QBC-ZKPAF performs exceptionally well across all attack categories, offering protection against replay, side-channel, collusion, MITM, and quantum computing attacks. Unlike other models, it effectively mitigates collusion attacks, which none of the other models prevents, and provides robust defence against MITM attacks, outperforming several alternatives. The QBC-ZKPAF is also resilient to quantum computing threats, ensuring its future-proof nature in the face of emerging computational challenges.

Table 3 provides a performance comparison of various cryptographic models, including TUVDS, HMAC, Digital Signature, UDVS, Ring Signature, and the proposed QBC-ZKPAF, focusing on key metrics such as overall throughput (MB/s), computational overhead (CPU %), computational speed (ms), computational efficiency (%), operational costs, and transaction/latency costs. It highlights the proposed QBC-ZKPAF model's superior performance in terms of computational efficiency (98%) and low computational overhead (15%), making it a highly effective and

TABLE 2. Robustness of proposed QBC-ZKPAF over other models with respect to potential attacks.

Attacks	TUVDS	HMAC	Digital Signature	UDVS	Ring Signature	Proposed QBC-ZKPAF
Replay Attacks	Yes	Yes	Yes	No	Yes	Yes
Side-Channel Attacks	No	No	Yes	Yes	Yes	Yes
Collusion Attacks	Yes	No	No	No	No	Yes
Man-in-the-Middle (MITM) Attacks	Yes	Yes	No	Yes	No	Yes
Quantum Computing Attacks	No	No	No	No	No	Yes
Quantum MITM	No	No	No	No	No	yes

TABLE 3. Performance of proposed QBC-ZKPAF over other models in terms of computational overhead, efficiency, and cost.

Model	Overall Throughput Overhead (MB/s)	Computational Overhead (CPU %)	Computational Speed (ms)	Computational Efficiency (%)	Operational Cost	Transaction and Latency Costs
TUVDS	150	80%	300	60%	High	High
HMAC	50	20%	50	95%	Moderate	High
Digital Signature	200	75%	500	70%	High	High
UDVS	100	50%	200	80%	Moderate	Low
Ring Signature	220	85%	700	55%	High	Moderate
Proposed QBC-ZKPAF	40	15%	40	98%	Low	Low

energy-efficient solution compared to other models. This comparison helps assess the practical benefits of different cryptographic mechanisms in terms of their impact on system performance and cost efficiency.

V. CONCLUSION

This research introduced QBC-ZKPAF, a Unified Quantum-Resilient Blockchain-ZKP Privacy Authentication Framework. This utilized blockchain technology combined with ZTA plus post-quantum cryptographic methods and hybrid RLBK for secured data transmission. The privacy-preserving signature generation and verification are supported by ZKP. The features established include traceability and auditability and offering tamper-proof records in the management of access and identities using blockchain’s decentralized and immutable ledger. Integrating blockchain with ZKP, therefore, balances privacy with accountability since only revealing tracing mechanisms under authorized conditions proves identities. The framework addresses the rising concern related to security and privacy in IoT environments, especially in an environment where emerging quantum threats have arrived. Decentralizing identity management, improving key generation, and preserving privacy during communication, QBC-ZKPAF, by its merits, marks as an innovative and effective cybersecurity solution for modern IoT networks that ensure user privacy with robust security across the system. Experimental results demonstrated that the achieved results of the proposed QBC-ZKPAF model reflect heavily improved security as well as performance, with a reduction of the authentication time to 100 ms and an increase in throughput to 700 TPS, which also develops 98% privacy preservation with an energy consumption level of 0.7 J and 0.98 quantum resilience; it is highly scalable, with fewer false acceptances of 0.005 and rejections of 0.05, and also has a 96% effect of access control, so it is very suitable for blockchain

applications and IoT networks that are sensitive to privacy. These enhancements besides demonstrating the better performance as well as scalability, energy efficiency, and security of the model render it highly suitable for IoT networks as well as privacy-sensitive blockchain applications. Future studies could focus on further optimizing the QBC-ZKPAF model for specific IoT sectors, such as healthcare and smart cities, to address unique device and network constraints. Additionally, research could explore its integration with emerging quantum-resistant cryptographic standards to future-proof the framework.

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